Yeojin Jung

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Education

Arkansas State University — Jonesboro, AR
B.S. in Computer Science, GPA: 3.87/4.00 — *Aug 2024 – Dec 2026 (Expected)*

Sangmyung University — Seoul, South Korea
B.S. Coursework in Human-Centered Artificial Intelligence, GPA: 4.29/4.50 — *Mar 2022 – Dec 2023*

Research Interests

Machine Learning, Deep Learning, Computer Vision, Video Understanding, Embodied AI

Research Experience

Undergraduate Research Assistant — Aug 2024 – May 2025, Feb 2026 – Present
Arkansas State University — *Jonesboro, AR*

- Conducted research on antimicrobial resistance prediction using bacterial whole-genome sequencing data.
- Developed and evaluated machine learning and deep learning models, including CNNs and Transformers, for antimicrobial resistance prediction.
- Designed genomic feature extraction, data preprocessing, and model evaluation pipelines.
- Evaluated model generalization using external bacterial genome datasets and performed statistical analysis and data visualization of experimental results.
- Extended binary AMR prediction from ACM-BCB Companion 2025 work into a multi-label framework for antibiotic prioritization, leading to an accepted regular paper at IEEE BIBM 2026.

Publications

Yeojin Jung, Chia Lee, and Donghoon Kim. "Multi-Label Antimicrobial Resistance Prediction for Antibiotic Prioritization." *2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026)*, Accepted as a Regular Paper, Dec 2026. **Acceptance rate: 19.48% (480/2,463).**

Yeojin Jung*, Namkyeong Kim*, and Donghoon Kim. "Improving Antimicrobial Resistance (AMR) Phenotype Prediction for Unseen Bacteria through Data Augmentation and Machine Learning." *ACM-BCB Companion Proceedings (ParBio Workshop)*, 2025.

* Equal contribution.

Projects

AI Closet – AI-Based Outfit Recommendation System | *Python, CLIP, ONNX* — Aug 2026 – Present

- Developing an AI-based wardrobe application that recommends outfits based on users' clothing images, weather conditions, and event context.
- Implemented zero-shot clothing image classification using CLIP and converted its image encoder to a quantized ONNX model for efficient on-device deployment.

AMR Prediction Web Application | *Python, PyTorch, Deep Learning* — Sep 2026 – Present

- Contributing to the development of a web-based antimicrobial resistance prediction system based on deep learning models developed through our AMR research.
- Developed and provided inference code to load trained models, preprocess genomic input data, and generate predictions for integration with the web application backend.

AWS & KT AICE Big Data AI Competition | *Python, Computer Vision* — May 2023 – Sep 2023

- Led a student team in developing a computer vision system for analyzing abnormal driving behavior from CCTV footage.
- Applied vehicle detection and tracking to extract and analyze vehicle trajectories for abnormal driving behavior detection.

Presentations

"Machine Learning-Based Prioritization of Antibiotic Susceptibility Testing Using Whole-Genome Data," CREATE@State Symposium, Arkansas State University, Apr 2026

"Using Machine Learning to Predict Antimicrobial Resistance (AMR) Phenotypes," CREATE@State Symposium, Arkansas State University, Apr 2025

"Using Machine Learning to Predict Antimicrobial Resistance (AMR) Phenotypes," STEM Posters at the Capitol, Little Rock, Arkansas, Jan 2025 (*Selected Presenter*).

Awards & Honors

Chancellor's List (4.0/4.0 GPA), Arkansas State University (Spring 2025, Fall 2025, Spring 2026)

3rd Place Poster Award, CREATE@State Symposium, Arkansas State University, Apr 2025

Selected Participant, 1st Korea–U.S. High-Tech Industry Student Exchange Program, Jun 2024

Grand Prize, SW-Centered University Generative AI Utilization Contest, Sangmyung University, Nov 2023

Top 7 Finalist, Big Data AI Competition with AWS and KT AICE, Sep 2023

Academic Achievement Award, Sangmyung University, Aug 2023

SW Scholarship (Ranked 2nd in Department; 2/64), Sangmyung University, Feb 2023

SW Scholarship (Ranked 1st in Department; 1/84), Sangmyung University, Aug 2022

Technical Skills

Programming Languages: Python, Java, C/C++, SQL, Kotlin
Machine Learning & Data Science: scikit-learn, Pandas, NumPy, Matplotlib
Deep Learning: PyTorch, TensorFlow
Tools & Platforms: Git, MySQL